SOMAIYA

VIDYAVIHAR

K J Somaiya Institute of Technology

An Autonomous Institute Permanently Affiliated to the University of Mumbai

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Class : Second Year (SY) 2026-27 **Department : Information Technology**

Course code: ITXS37 **Subject: DSA**

Experiment No:2

Aim: Experiment No. 01 (Implementation of Stack using Array)

Program

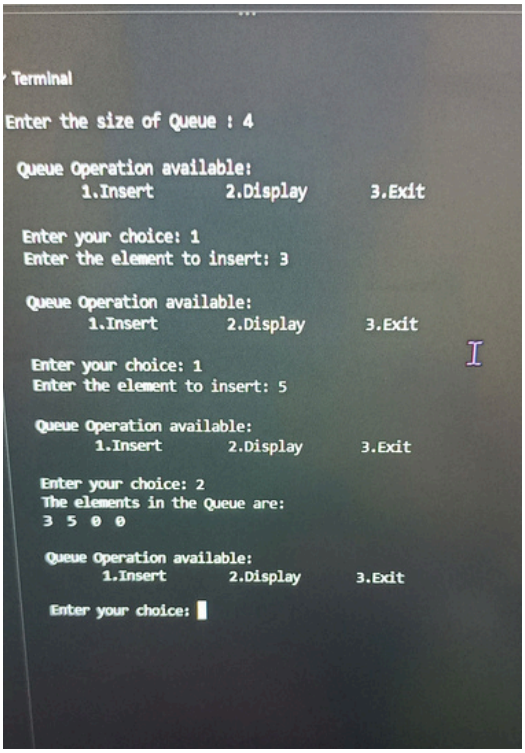
```
#include <stdio.h>
int Q[10], f = -1, R = -1, i, n, x, choice;
void insert();
void display();
void main()
{
   printf("Enter the size of Queue : ");
   scanf("%d", &n);
   do
   {
      printf("\n Queue Operation available: \n");
      printf("\t1.Insert \t2.Display \t3.Exit \n");
      printf("\n Enter your choice: ");
```

```
        scanf("%d", &choice);
        switch (choice)
        {
        case 1:
           insert();
           break;
        case 2:
           display();
           break;
        case 3:
           printf("Exit: Program Finished !! ");
           break;
        default:
           printf("Please enter a valid choice 1, 2, 3, 4 \n");
           break;
        }
    } while (choice != 4);
}
void insert()
{
    if (R >= n - 1)
    {
        printf(" Queue Overflow ! \n");
    }
    else
```

```
    {
        printf(" Enter the element to insert: ");
        scanf("%d", &x);
        R++;
        Q[R] = x;
        if (f == -1)
        {
            f = 0;
        }
    }}
void display()
{
    if (R < 0)
    {
        printf(" Queue is empty ! \n");
    }
    else
    {
        printf(" The elements in the Queue are: \n");
        for (i = f; i < n; i++)
        {
            printf(" %d ", Q[i]);
        }
        printf("\n");
    }
```

}

Outcome:


Terminal
Enter the size of Queue : 4
Queue Operation available:
1.Insert 2.Display 3.Exit
Enter your choice: 1
Enter the element to insert: 3
Queue Operation available:
1.Insert 2.Display 3.Exit
Enter your choice: 1
Enter the element to insert: 5
Queue Operation available:
1.Insert 2.Display 3.Exit
Enter your choice: 2
The elements in the Queue are:
3 5 0 0
Queue Operation available:
1.Insert 2.Display 3.Exit
Enter your choice:

Conclusion: SUCESSFULLY EXCUTED THE STACK USING STRING